

Hermes Agent — Purchase Order Intake & Order-Tracking Workflow

Context

Customer purchase orders (POs) currently land in several different salespeople's inboxes. With no central intake there is no reliable view of the open / pending / closed backlog, and orders get lost. The **"Hermes" agent** will:

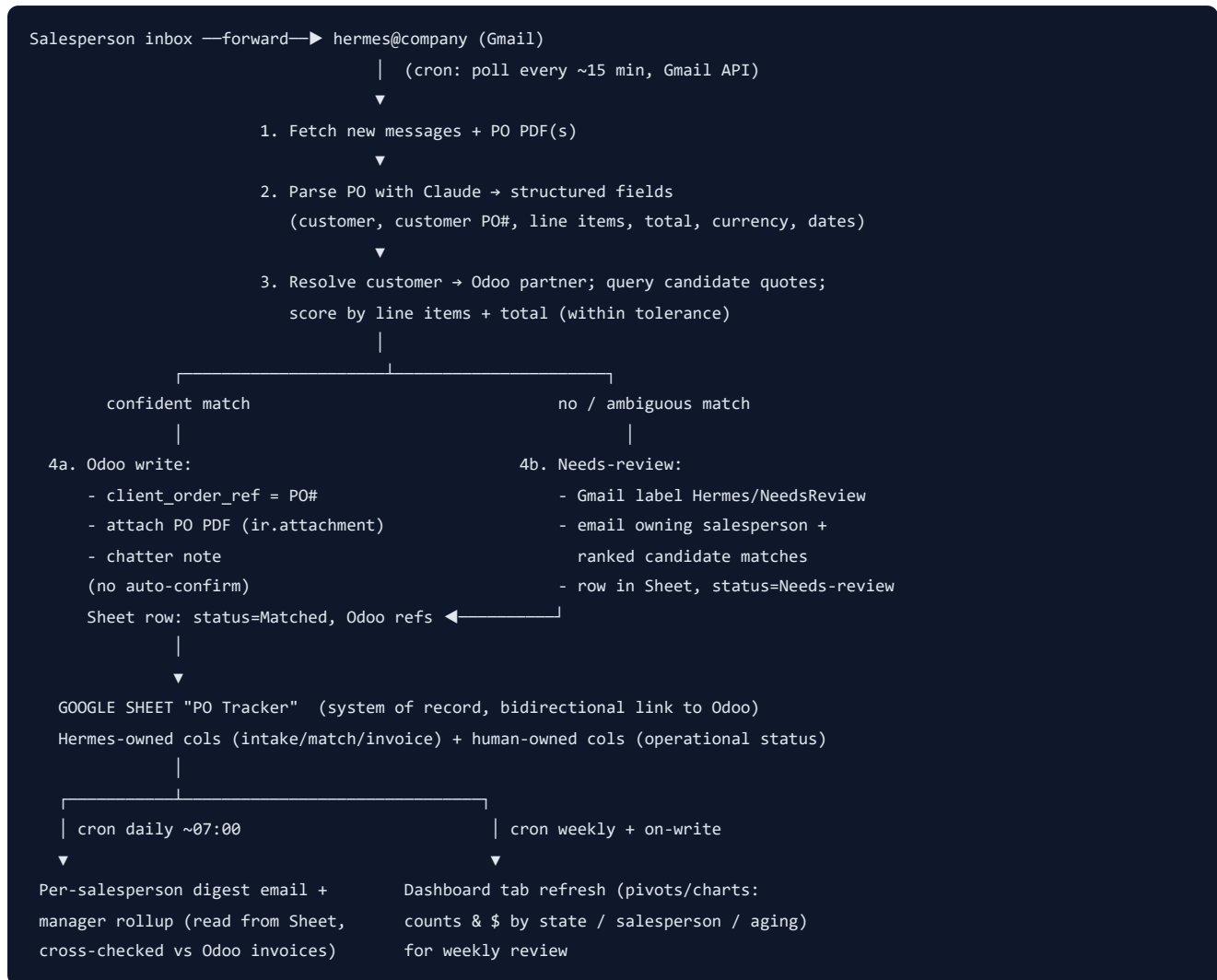
1. Receive forwarded customer PO emails at one dedicated mailbox.
2. Parse each PO and **match it to the corresponding quotation in Odoo**, then write the customer's PO number back into the Odoo Sales Order and attach the PO PDF.
3. Build and maintain a **dedicated PO-tracking database (a Google Sheet)** as the system of record for order status, **bidirectionally linked to Odoo**.
4. **Email each salesperson a daily digest** of their orders, and maintain a ****weekly- review dashboard**** off the tracker.

Key design decision (revised per user): Odoo is used narrowly — for **quotes** (to match against) and **invoices**. The **Google Sheet PO Tracker is the source of truth** for order status. Hermes links each PO to Odoo and vice-versa, and never lets the two drift by owning a clear set of columns while leaving operational status to humans.

Confirmed requirements (from clarifying questions)

- **Email:** Google Workspace (Gmail); salespeople forward POs to a Hermes mailbox.
 - **Odoo:** Odoo.sh (full External API, staging branches). Used for quotes + invoices only.
 - **Tracker / dashboard:** **Google Sheets** (DB + weekly-review dashboard in one).
 - **Runtime:** Hermes = Claude agent **skills** + **scheduled (cron) routines**, backed by thin Python connectors for the deterministic Gmail / Odoo / Sheets I/O.
 - **Matching signals:** customer identity + customer's PO number + line items + total.
 - **On a confident match:** write PO# into the Odoo SO (`client_order_ref`), attach the PO PDF, post a chatter note. **Do NOT auto-confirm** the SO.
 - **On no/ambiguous match:** Needs-review bucket + email the owning salesperson.
 - **Lifecycle (granular):** Received → Matched → In production → Shipped → Invoiced → Closed (plus Needs-review). Some auto, some human-set (see table).
 - **Closing rule: hybrid** — Hermes detects the Odoo invoice and suggests Invoiced/Closed; a person confirms.
 - **Reminders:** daily email digest per salesperson + manager rollup; weekly dashboard review.
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Architecture & Data Flow



Order lifecycle (states & who sets them)

State	Set by	Trigger
Needs-review	Hermes (auto)	PO parsed but no/ambiguous Odoo quote match
Received	Hermes (auto)	PO email parsed and logged to the Sheet
Matched	Hermes (auto)	Confident match; PO# + PDF written to Odoo SO, refs linked both ways
In production	Human	Salesperson updates the Status dropdown in the Sheet
Shipped	Human	Salesperson updates the Status dropdown in the Sheet
Invoiced	Hermes suggests → Human confirms	Linked Odoo SO becomes invoiced; Hermes flags, person confirms
Closed	Human confirms	After invoice (optionally paid) confirmed

Hermes only writes the auto states and the *suggestion* flag for Invoiced; it never overwrites a human-owned status cell.

Google Sheet "PO Tracker" — schema

Two column groups so Hermes and humans never clobber each other:

Hermes-owned (the agent writes; humans don't edit):

`PO_ID` (internal key) · `Received_At` · `Salesperson` · `Customer` · `Customer_PO#` ·
`Line_Summary` · `Total` · `Currency` · `Match_Status` · `Match_Confidence` ·
`Odoo_Partner_ID` · `Odoo_Quote/SO_ID` · `Odoo_SO_Number` · `Odoo_Invoice_ID` ·
`PO_PDF_Link` · `Invoiced_Suggested` (bool) · `Last_Synced_At`

Human-owned (data-validation dropdowns / free text):

`Status` (lifecycle dropdown) · `Notes` · `ETA` · `Close_Confirmed` (checkbox)

Tabs: `Orders` (the rows above), `Dashboard` (pivots/charts for weekly review),

`Salespeople` (Gmail ↔ Odoo user ↔ manager map), `Audit` (match/decision log).

Components (files to create)

```
hermes/  
  config/  
    hermes.config.yaml      # "Configuration C" – see schema below  
  connectors/  
    gmail_client.py         # Gmail API: list/fetch/label messages, send mail  
    odoo_client.py          # Odoo External API (xmlrpc): quotes, sale.order writes, attachments, chatter, invoice status  
    sheets_client.py        # Google Sheets API: read/append/update Orders, refresh Dashboard (respects column ownership)  
    po_parser.py            # Claude call: PDF/email → structured PO (Anthropic SDK document blocks; vision/OCR fallback)  
  core/  
    matcher.py              # partner resolve + line/total scoring + threshold  
    tracker.py              # lifecycle rules, idempotency, Sheet row upsert (Hermes-owned cols only)  
    digest.py               # per-salesperson + manager summaries; dashboard pivots  
  skills/  
    process-po-inbox/       # Claude skill: orchestrates intake (steps 1-4)  
    daily-order-digest/     # Claude skill: builds + sends digests, refreshes invoice suggestions  
  scripts/  
    run_intake.py           # entrypoint for the intake cron/skill  
    run_digest.py           # entrypoint for the digest + dashboard cron/skill  
  .env                     # secrets (NOT committed)  
  requirements.txt
```

Division of labor: **Python connectors** do deterministic API I/O; **Claude (skill)** does

the judgment — parsing the PO, fuzzy partner/quote matching, drafting notes and emails.

"Configuration C" — schema (`hermes.config.yaml`)

```
gmail:
  account: hermes@company.com
  credentials_path: ../secrets/google_oauth.json
  poll_query: "is:unread -label:Hermes/Processed"
  labels: { processed: Hermes/Processed, needs_review: Hermes/NeedsReview }
odoo:
  url: https://yourco.odoo.com
  db: yourco-main
  username: hermes-bot@company.com
  api_key_env: OD00_API_KEY
sheets:
  spreadsheet_id: <PO-Tracker sheet id>
  tabs: { orders: Orders, dashboard: Dashboard, people: Salespeople, audit: Audit }
  credentials_path: ../secrets/google_service_account.json
matching:
  candidate_states: [draft, sent]
  lookback_days: 120
  amount_tolerance_pct: 2.0
  compare: untaxed_total
  confidence_threshold: 0.85          # >= -> confident; else Needs-review
write:
  # Odoo write on confident match
  customer_ref_field: client_order_ref # Odoo "Customer Reference" = customer PO#
  attach_pdf: true
  post_chatter_note: true
  auto_confirm_so: false             # annotate only
invoice:
  detect_from: invoice_status        # poll Odoo; suggest Invoiced when SO is invoiced
  require_human_confirm: true         # hybrid close
reminders:
  daily_digest_time: "07:00"
  timezone: America/Chicago
  manager_rollup_to: [manager@company.com]
  weekly_dashboard_refresh: "Mon 06:30"
```

Setup prerequisites (one-time)

Google / Gmail / Sheets

- Dedicated Hermes mailbox (e.g. `hermes@company.com`).
- Google Cloud project → enable **Gmail API** + **Sheets API**. OAuth client (or service account w/ domain-wide delegation) for the mailbox; a **service account** shared onto the PO Tracker sheet for Sheets writes.
- Create the **PO Tracker** Google Sheet with the tabs/columns above (Hermes can scaffold it).
- Have salespeople **auto-forward** customer PO emails to the Hermes address (per-person Gmail filter so they don't have to remember).

Odoo.sh

- Dedicated least-privilege Odoo user **"Hermes"** (read quotes, write `sale.order` `client_order_ref`, create `ir.attachment`, post messages, read `invoice_status`). Generate an **API key**. Validate writes on an **Odoo.sh staging branch** first.

Runtime

- Python env: `google-api-python-client`, `google-auth(-oauthlib)`, `anthropic`, `pyyaml`; Odoo via stdlib `xmlrpc.client`.
- Secrets in `.env` / secret store — never committed.
- Register two scheduled routines (intake ~15 min; digest+dashboard daily/weekly) via the Claude Code scheduler, each wired to its skill + endpoint.

Implementation milestones

1. **Connectors + config** — Gmail, Odoo (read-only first), Sheets; prove auth and scaffold the PO Tracker tabs/columns.
 2. **PO parser** — forwarded email + PDF → structured PO; inline vs attached PDFs; scanned/ image fallback via vision.
 3. **Matcher** — partner resolution (exact + fuzzy + alias map) + line/total scoring within tolerance → confidence + audit trail.
 4. **Tracker upsert** — write Received/Matched rows (Hermes-owned cols only); idempotency on Gmail message ID + PO#; never touch human-owned cols.
 5. **Odoo write path** — confident match: `client_order_ref`, attach PDF, chatter note; store Odoo refs back on the Sheet row. Validate on staging. No auto-confirm.
 6. **Needs-review + notifications** — Gmail label, email owning salesperson w/ ranked candidates, Needs-review row.
 7. **Invoice sync (hybrid close)** — poll Odoo `invoice_status`; set `Invoiced_Suggested`, notify; human confirms via the Sheet to reach Invoiced/Closed.
 8. **Digest + dashboard** — daily per-salesperson + manager rollup; weekly Dashboard tab pivots/charts.
 9. **Schedule + harden** — register cron; dry-run mode, logging, retries, failure alerts.
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Risks & mitigations

- **Customer name ≠ Odoo partner name** → fuzzy match + alias map; unresolved → Needs-review.
 - **Total mismatch** (tax/shipping/discount/currency) → compare untaxed totals + tolerance; show the delta in the review notice.
 - **Multiple similar open quotes** → ambiguous → Needs-review with ranked candidates.
 - **Scanned/image POs** → Claude vision/OCR; low confidence → Needs-review.
 - **Sheet write conflicts** → strict column ownership; Hermes only upserts its own columns and matches rows by `PO_ID`.
 - **Duplicate forwards** → idempotency on Gmail message ID + PO# in the Audit tab.
 - **Financial safety** → Hermes annotates only; humans confirm both the SO and the close.
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Verification (end-to-end)

- **Dry-run:** process a real sample PO; log parsed fields + candidate matches + intended action, writing nothing.
 - **Odoo staging writes:** run the write path on the Odoo.sh staging branch; confirm `client_order_ref`, attached PDF, and chatter note appear; re-run same message → no dupes.
 - **Sheet round-trip:** confirm a Received/Matched row appears with Odoo refs; manually edit a human-owned cell and re-run → Hermes does not overwrite it.
 - **Needs-review:** feed a PO with no match → Gmail label + salesperson email + Needs-review row.
 - **Invoice/close:** mark the linked SO invoiced in staging → `Invoiced_Suggested` set + notice; confirm in the Sheet → reaches Closed.
 - **Digest + dashboard:** point manager rollup at yourself; verify buckets/totals vs the Sheet, and the Dashboard tab pivots, then enable schedules.
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Assumptions / defaults (tell me to change any)

- Operational states (In production, Shipped) and the final Close are updated by humans via dropdowns in the Sheet; Hermes only sets auto states + the Invoiced suggestion.
 - Customer PO# goes in Odoo's `client_order_ref` (searchable) plus a chatter note.
 - Dedicated least-privilege Odoo + Google service accounts; Hermes never confirms SOs.
 - Intake polls every ~15 min (cron); upgradeable to Gmail push later.
 - Daily digest is email; weekly review is the Sheet Dashboard tab (Slack/Teams later if wanted).
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